

# General Notes on FEOM

Cole Hunt

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This PDF contains general notes on the FEOM project, including information about the project structure, coding conventions, and other relevant details.

## I. Project Structure

This project implements HEOM using FORTRAN, and implements basic HEOM with the standard K and L cutoffs. In the notation used by FORTRAN propagation, we have

$$\begin{aligned} \frac{d}{dt}\rho_{\mathbf{n}}(t) = & -\frac{i}{\hbar}H_s^\times\rho_{\mathbf{n}}(t) - \sum_k n_k\gamma_k\rho_{\mathbf{n}}(t) - \frac{i}{\hbar}\sum_k \left(c_U(k, n_k)V^\times\rho_{\mathbf{n}_k^+}(t)\right) \\ & + \frac{i}{\hbar}\sum_k \left(c_{\text{DLEFT}}(k, n_k)V^L\rho_{\mathbf{n}_k^-}(t) + c_{\text{DRIGHT}}(k, n_k)V^R\rho_{\mathbf{n}_k^-}(t)\right), \end{aligned} \quad (1)$$

where  $\rho_{\mathbf{n}}(t)$  is the full set of ADOs,  $\mathbf{n}$  is the vector of ADO indices  $\{n_k\}$ , (with  $\mathbf{n}_k^\pm$  denoting the same set of indices, but with  $n_k \pm 1$ ). We use superoperator notation throughout, and define our coefficients  $\{\gamma_k, c_U, c_D\}$  differently for each version of the HEOM used.

### A. Shi/Chen scaled HEOM

For Shi/Chen scaled HEOM, the equations of motion are given by

$$\frac{d}{dt}\rho_{\mathbf{n}}(t) = -\frac{i}{\hbar}H_s^\times\rho_{\mathbf{n}}(t) - \sum_k n_k\gamma_k\rho_{\mathbf{n}}(t) - \frac{i}{\hbar}\sum_k \left(\sqrt{n_k+1}\mathcal{L}_{k-}\rho_{\mathbf{n}_k^+}(t) + \sqrt{n_k}\mathcal{L}_{k+}\rho_{\mathbf{n}_k^-}(t)\right), \quad (2)$$

where  $\rho_{\mathbf{n}}(t)$  is the full set of ADOs,  $\mathbf{n}$  is the vector of ADO indices  $\{n_k\}$ , (with  $\mathbf{n}_k^\pm$  denoting the same set of indices, but with  $n_k \pm 1$ ). We use superoperator notation throughout,  $H_s$  as the system Hamiltonian, the superoperators  $\mathcal{L}_{k-}$  and  $\mathcal{L}_{k+}$  are defined as the superoperator

$$\mathcal{L}_{k-}\cdot = \sqrt{|C_k|}V^\times, \quad \mathcal{L}_{k+}\cdot = \frac{1}{\sqrt{|C_k|}}(C_kV^L - C_k^*V^R), \quad (3)$$

where  $V$  is the system-bath coupling operator,  $C_k$  is the coupling strength for the  $k$ 'th bath mode, and the constants  $C_k$  and  $\gamma_k$  are obtained from the bath correlation function (BCF) as

$$C(t) = \sum_k C_k e^{-\gamma_k t}, \quad (4)$$

which results in the coefficients,

$$c_U(k, n_k) = \sqrt{n_k+1}|C_k|, \quad c_{\text{DLEFT}}(k, n_k) = -\sqrt{\frac{n_k}{|C_k|}}C_k, \quad c_{\text{DRIGHT}}(k, n_k) = \sqrt{\frac{n_k}{|C_k|}}C_k^*, \quad (5)$$

and  $\gamma_k$  being the decay rate of the  $k$ 'th bath mode, and  $K$  modes chosen to represent the BCF. In what follows, any sums over  $k$  are assumed to go from 1 to  $K$ , unless otherwise specified.

### B. Free-Pole HEOM

For Free-Pole HEOM, the bath correlation function is represented in terms of a sum of damped exponentials,

$$C_k = \sum_{k=1}^K d_k e^{-z_k t}, \quad (6)$$

with  $d_k, z_k \in \mathbb{C}$ . the equations of motion are given by

$$\begin{aligned} \frac{d}{dt} \rho_{\mathbf{m}, \mathbf{l}}(t) = & -\frac{i}{\hbar} H_s^\times \rho_{\mathbf{m}, \mathbf{l}}(t) - \left( \sum_k m_k z_k + l_k z_k^* \right) \rho_{\mathbf{m}, \mathbf{l}}(t) - \frac{i}{\hbar} \sum_k \sqrt{(m_k + 1) d_k} V^\times \rho_{\mathbf{m}_k^+, \mathbf{l}}(t) \\ & - \frac{i}{\hbar} \sum_k \sqrt{(l_k + 1) d_k^*} V^\times \rho_{\mathbf{m}, \mathbf{l}_k^+}(t) - \frac{i}{\hbar} \sum_k \sqrt{m_k d_k} V^L \rho_{\mathbf{m}_k^-, \mathbf{l}}(t) + \frac{i}{\hbar} \sum_k \sqrt{l_k d_k^*} V^R \rho_{\mathbf{m}, \mathbf{l}_k^-}(t). \end{aligned} \quad (7)$$

We can combine the two ADO intices (there are two of them due to the forward and back paths being separated out in the influence functional) into a single vector  $\mathbf{n} = \{\mathbf{m}, \mathbf{l}\}$ , such that the EOM are now in the same form as the Shi/Chen scaled HEOM, with the coefficients defined as

$$c_U(k, n_k) = \begin{cases} \sqrt{(n_k + 1) d_k}, & k \leq K, \\ \sqrt{(n_k + 1) d_{k-K}^*}, & k > K \end{cases}, \quad c_{\text{DLEFT}}(k, n_k) = \begin{cases} -\sqrt{n_k d_k}, & k \leq K, \\ 0, & k > K \end{cases} \quad (8)$$

$$c_{\text{DRIGHT}}(k, n_k) = \begin{cases} 0, & k \leq K, \\ \sqrt{n_k d_{k-K}^*}, & k > K \end{cases}, \quad \gamma_k = \begin{cases} z_k, & k \leq K, \\ z_{k-K}^*, & k > K \end{cases}. \quad (9)$$

## II. Bath conventions

In this section, we discuss the conventions used for the naming of the baths. We define the spectral density

$$J(\omega) = \frac{\pi}{2} \sum_\alpha \frac{c_\alpha^2}{m_\alpha \omega} [\delta(\omega - \omega_\alpha) + \delta(\omega + \omega_\alpha)], \quad (10)$$

where  $c_\alpha$  is the coupling strength,  $m_\alpha$  is the mass, and  $\omega_\alpha$  is the frequency of the  $\alpha$ 'th bath mode. It is assumed that  $\omega_\alpha \geq 0$ , such that  $J(\omega) = -J(-\omega)$ . With this definition, the equilibrium bath correlation function (BCF), at inverse temperature  $\beta$ , is given as

$$C(t) = \frac{\hbar}{2\pi} \int_{-\infty}^{\infty} d\omega J(\omega) \left( \coth \left( \frac{\beta \hbar \omega}{2} \right) \cos \omega t - i \sin \omega t \right), \quad (11)$$

which is done using residues. Now to get onto specific bath distributions.

### A. Debye bath

The Debye bath is defined by the spectral density

$$J(\omega) = \frac{\eta \gamma \omega}{\omega^2 + \gamma^2}, \quad (12)$$

where  $\eta$  is the coupling strength and  $\gamma$  is the cutoff frequency. The correlation function is of exponential form, parameterized by frequencies  $\{\gamma_k\}$ ,

$$\gamma_0 = \gamma, \quad \gamma_k = \omega_{n=k} = \frac{2n\pi}{\beta \hbar}, \quad (13)$$

and prefactors  $\{C_k\}$ , which are given by

$$C_0 = \frac{\hbar \eta \gamma}{2} \left( \cot \frac{\beta \hbar \gamma}{2} - i \right), \quad C_{n=k} = -\frac{2\eta \gamma}{\beta} \frac{\omega_n}{\gamma^2 - \omega_n^2}, \quad \forall k > 0. \quad (14)$$

## III. Truncations

There are a couple ways to truncate the HEOM, so it's good to write these down. The truncation falls into two categories, which I will call exponential term truncation and heirarchy truncation. The exponential term truncation is the approximation of the bath correlation function (BCF) by a finite number of exponentials. The heirarchy truncation is the removal of the dependence of the ADOs on those with a larger depth.

### A. Matsubara series truncation

This truncation was created by Ishizaki and Tanimura in Ref.[1], and is switched on by the variable “*lowTcoef\_switch*”. It is a Markovian approximation of the higher order Matsubara terms in the BCF;

$$e^{-\gamma_k t} \approx \frac{1}{\gamma_k} \delta_+(t), \quad \forall k > k_c, \quad (15)$$

where  $k_c$  is the cutoff in number of exponentials in the BCF<sup>1</sup> and  $\delta_+(t)$  is a delta function defined for  $t \geq 0$ . This cutoff  $k_v = M + 1$  for the example of a Debye bath, wherein there is a single exponential term, and  $M$  is the number of Matsubara terms retained. This approximation is inserted into the influence functional, allowing us to do the inner integral (over the history), giving another term in the dynamics,

$$\Xi \rho_{\mathbf{n}}(t) = \sum_{k > k_c} \frac{1}{\gamma_k} \mathcal{L}_k - \mathcal{L}_{k+} \rho_{\mathbf{n}}(t). \quad (16)$$

For the example of a Debye bath, we can do the sum formally. Due to all of the  $C_k$  for  $k > 0$  being real, we can write the superoperator as

$$\Xi_{\text{debye}} = -\frac{1}{\hbar^2} \sum_{k > k_c} \frac{1}{\gamma_k} C_k (V^\times)^2 = \frac{2\eta\gamma}{\beta\hbar^2} \sum_{n > k_c} \frac{1}{\gamma^2 - \omega_n^2} (V^\times)^2 \quad (17)$$

Then we use the fact that

$$\begin{aligned} \cot z &= \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2\pi^2} \\ \therefore \sum_{n=1}^{\infty} \frac{1}{\gamma^2 - \omega_n^2} &= \frac{\beta\hbar}{4\gamma} \left( \cot \frac{\beta\hbar\gamma}{2} - \frac{2}{\beta\hbar\gamma} \right), \end{aligned} \quad (18)$$

to calculate the sum, giving us

$$\Xi_{\text{debye}} = \eta \underbrace{\left[ \frac{1}{2\hbar} \left( \cot \frac{\beta\hbar\gamma}{2} - \frac{2}{\beta\hbar\gamma} \right) - \frac{2\gamma}{\beta\hbar^2} \sum_{n=1}^{k_c} \frac{1}{\gamma^2 - \omega_n^2} \right]}_{\text{lowTcoef}} (V^\times)^2. \quad (19)$$

## References

- [1] Akihito Ishizaki and Yoshitaka Tanimura. Quantum Dynamics of System Strongly Coupled to Low-Temperature Colored Noise Bath: Reduced Hierarchy Equations Approach. *Journal of the Physical Society of Japan*, 74(12):3131–3134, December 2005.

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<sup>1</sup>these are ordered in increasing real part, as then the approximation is more accurate for larger  $k$